Name : Utsav Golani
Roll No : 2401032
Batch : T12
Subject ASD&D

EXPERIMENT - 08

Aim: CI/CD Pipeline to host a website on AWS.

LO4, LO6 mapped:

LO4:: Containerize, orchestrate, and deploy applications using Docker, Docker Compose, Kubernetes, and cloud infrastructure tools.

LO6: Develop and demonstrate an end-to-end DevOps pipeline mini-project integrating all stages of DevOps from code commit to deployment and monitoring.

Theory:

CI/CD is a DevOps practice that automates the process of developing, testing, and deploying software applications. It helps developers deliver updates quickly, efficiently, and reliably.
Continuous Integration (CI) is the process of automatically integrating code changes into a shared repository, followed by automated building and testing.
Continuous Deployment (CD) is the process of automatically deploying successfully tested code to a production environment without manual intervention.
AWS provides various cloud services that help developers create automated CI/CD pipelines for hosting websites.

AWS Services Used

1. AWS CodePipeline: Automates the software release process, from source code to deployment.
2. AWS CodeBuild: Compiles source code, runs tests, and generates deployment artifacts.
3. AWS CodeDeploy: Automates application deployment to supported compute services such as Amazon EC2.
4. Amazon S3: Stores and hosts static website files such as HTML, CSS, and JavaScript.
5. Amazon EC2: Provides virtual servers for hosting dynamic websites and applications.
6. AWS IAM: Manages user permissions and secure access to AWS resources.
7. GitHub: Stores website source code and tracks changes using version control. **Working of the CI/CD Pipeline**

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The CI/CD pipeline follows these steps:

Step 1: Source Code Management

Developers write website code using HTML, CSS, JavaScript, or other technologies and upload it to a GitHub repository.

Step 2: Continuous Integration

Whenever changes are pushed to GitHub, AWS CodePipeline detects the updates and automatically starts the pipeline.

Step 3: Build Process

AWS CodeBuild retrieves the latest source code, installs required dependencies, runs tests, and prepares the website files for deployment.

Step 4: Continuous Deployment

After successful testing, the pipeline automatically deploys the website to an AWS hosting service such as Amazon S3 for static websites or Amazon EC2 for server-based applications.

Step 5: Website Hosting

The deployed website becomes accessible through its configured website URL or domain name.

Step 6: Automatic Updates

Whenever developers push new changes to GitHub, the pipeline repeats the process and updates the hosted website automatically.

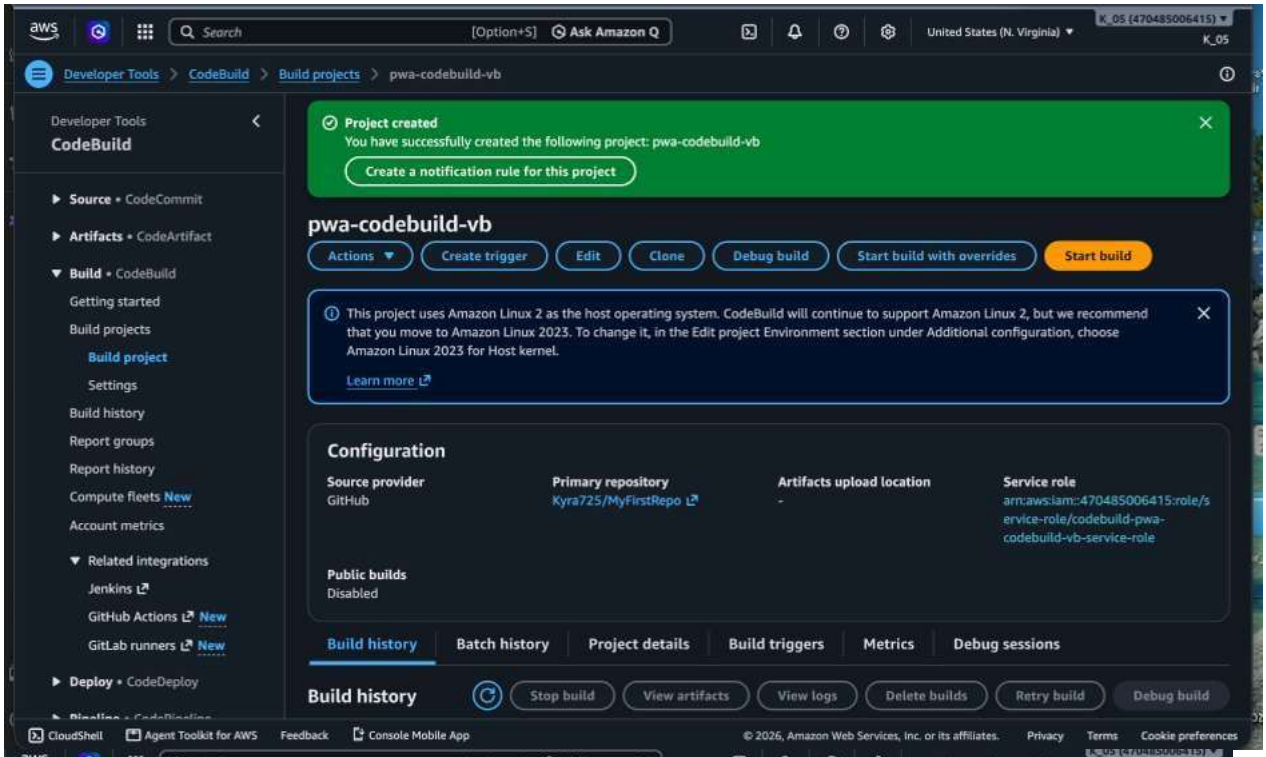
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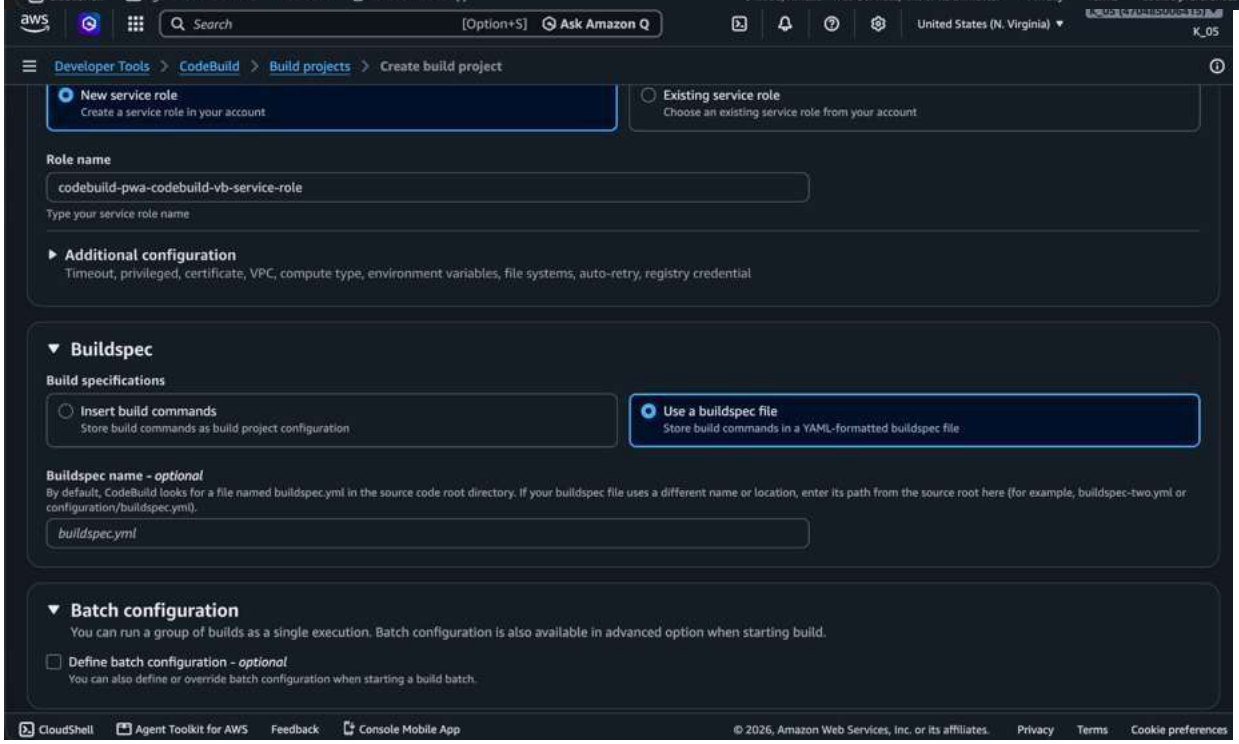
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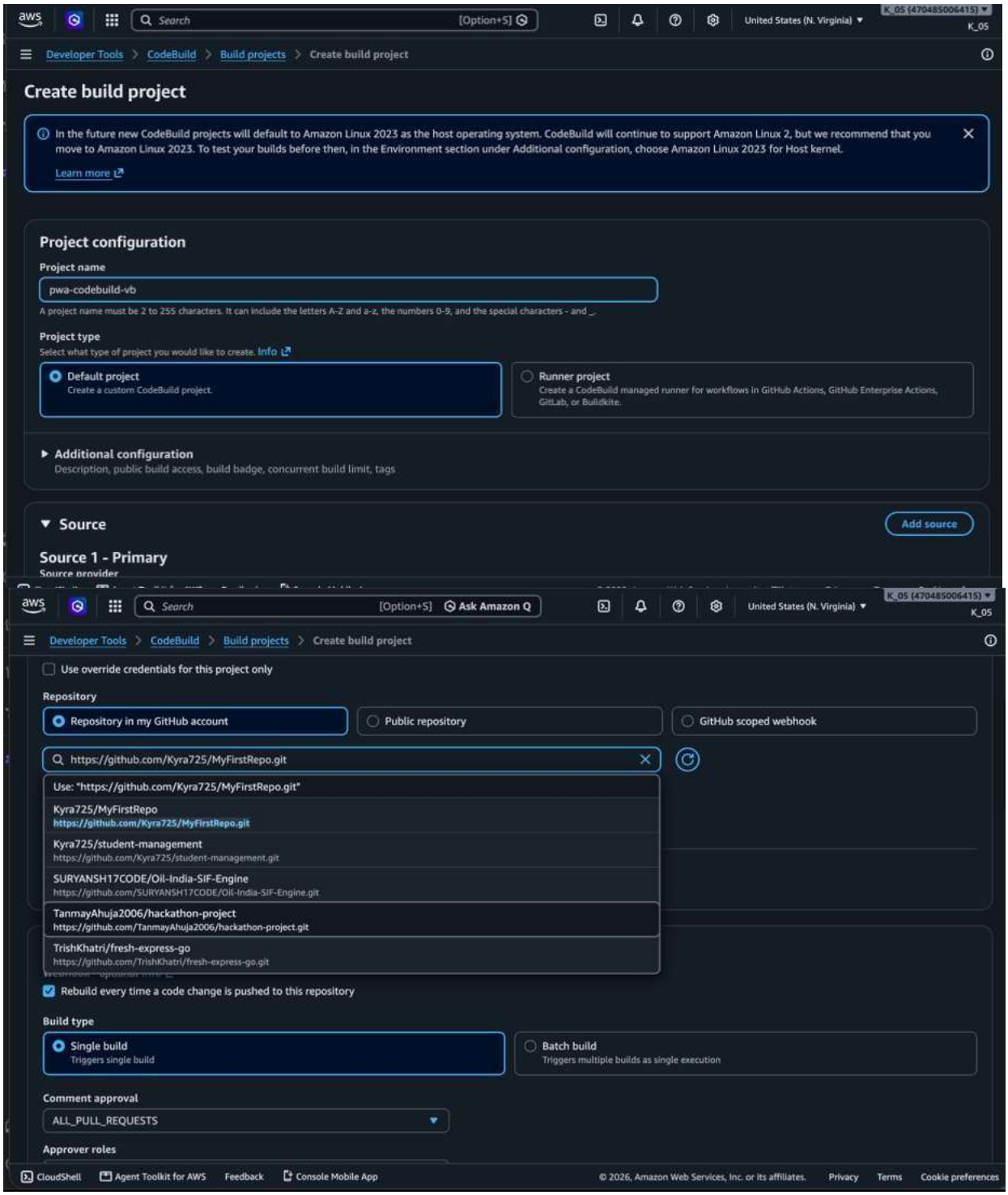


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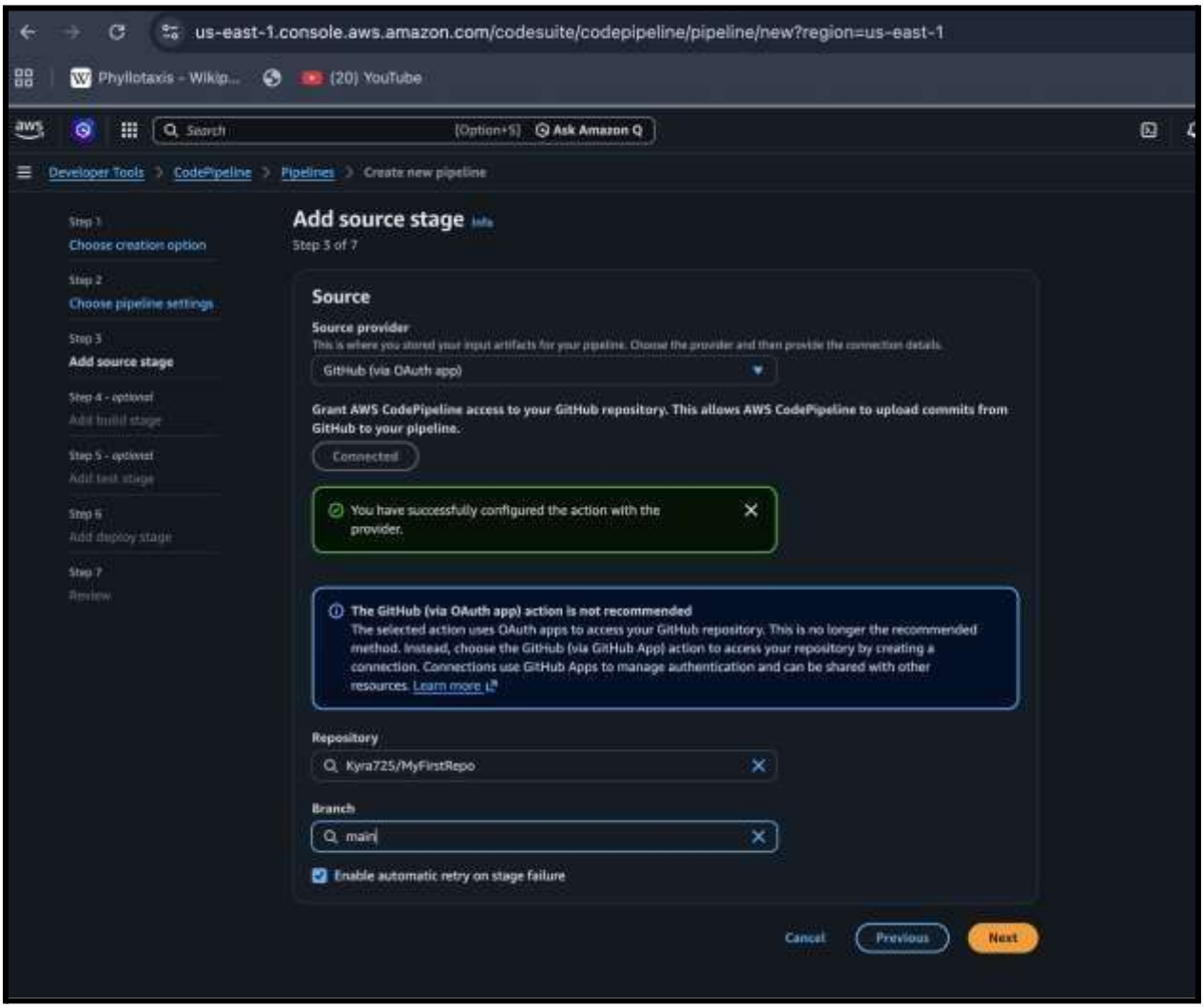


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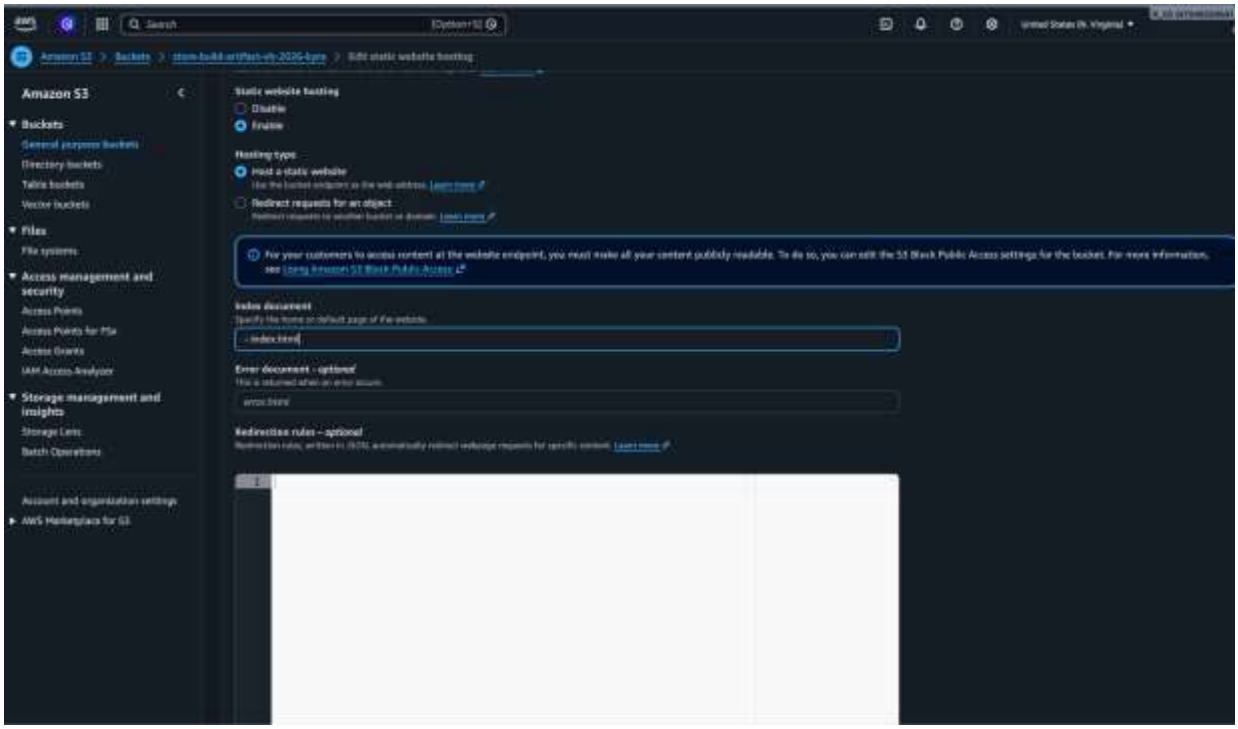
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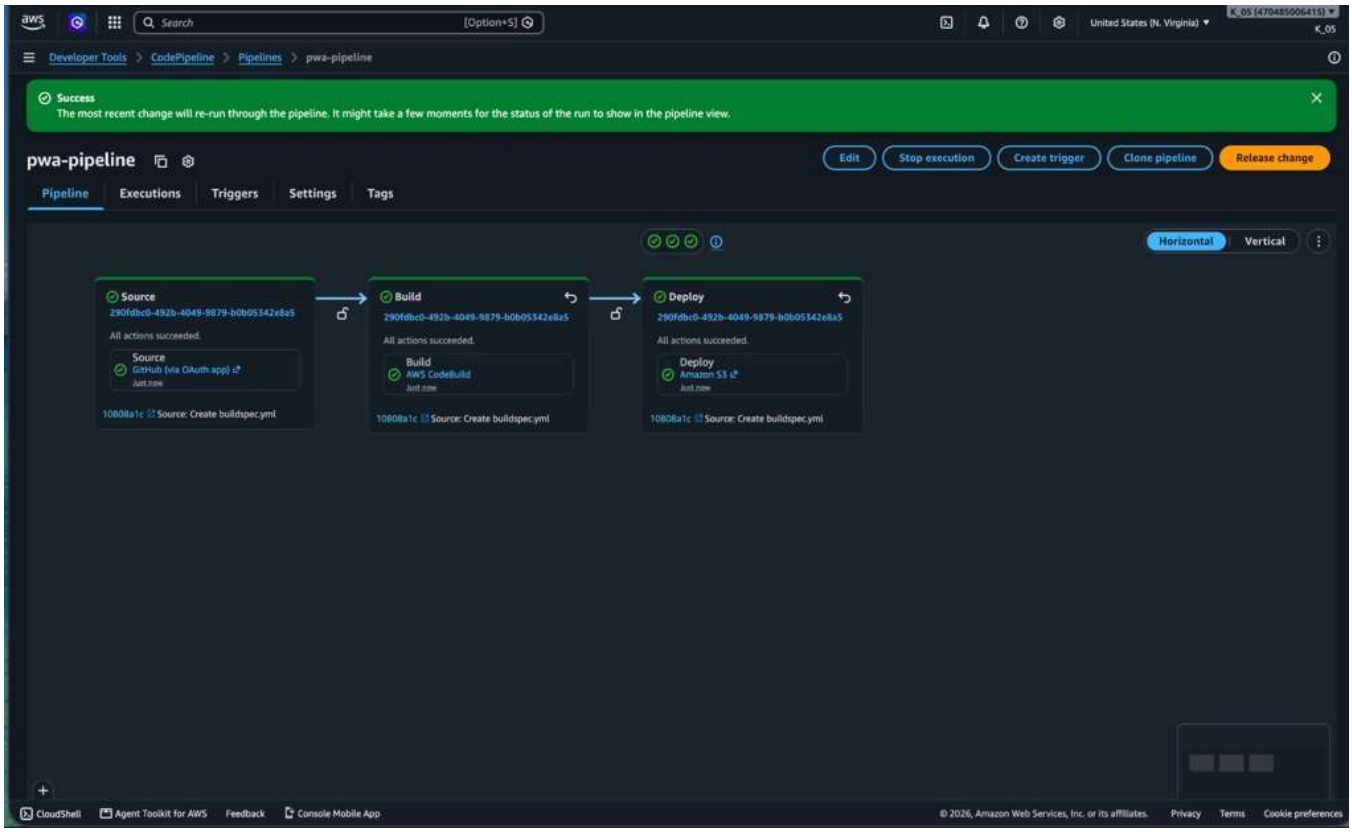


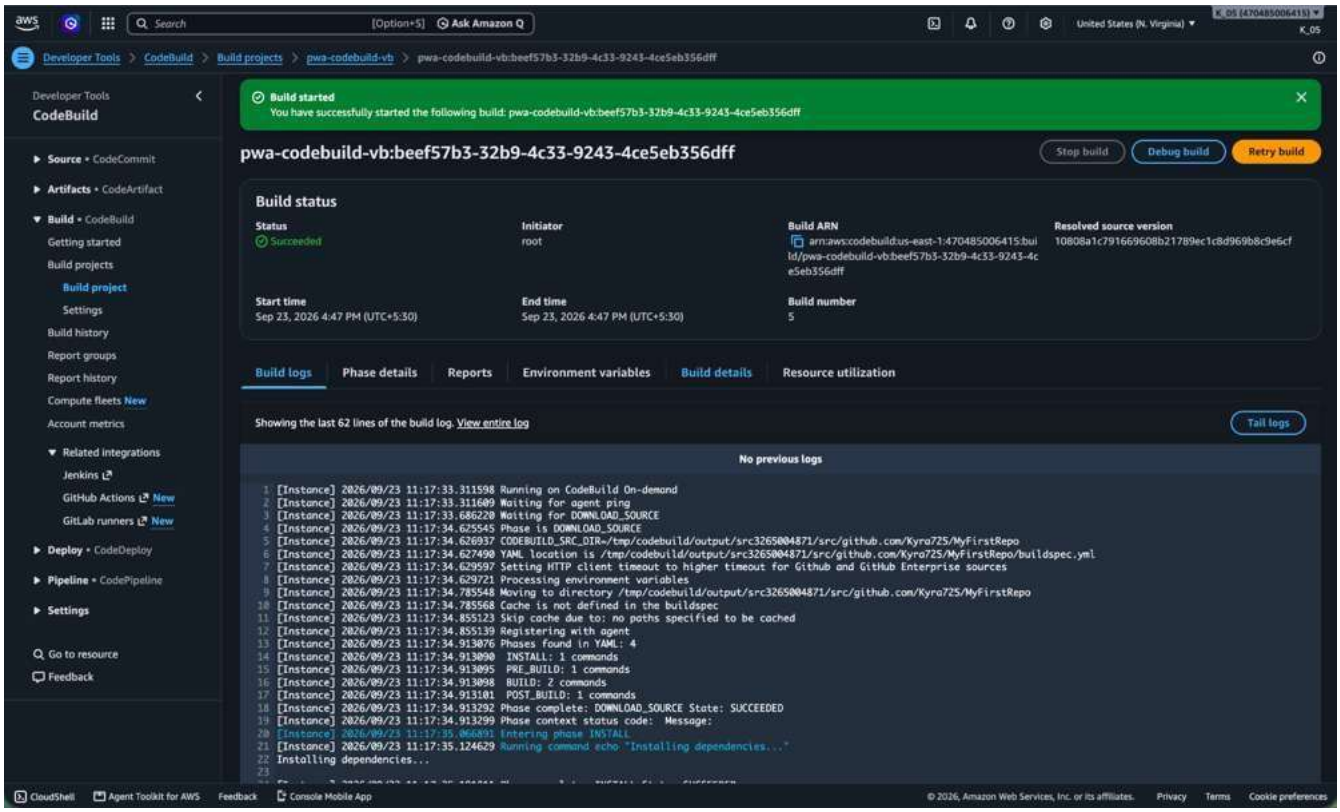
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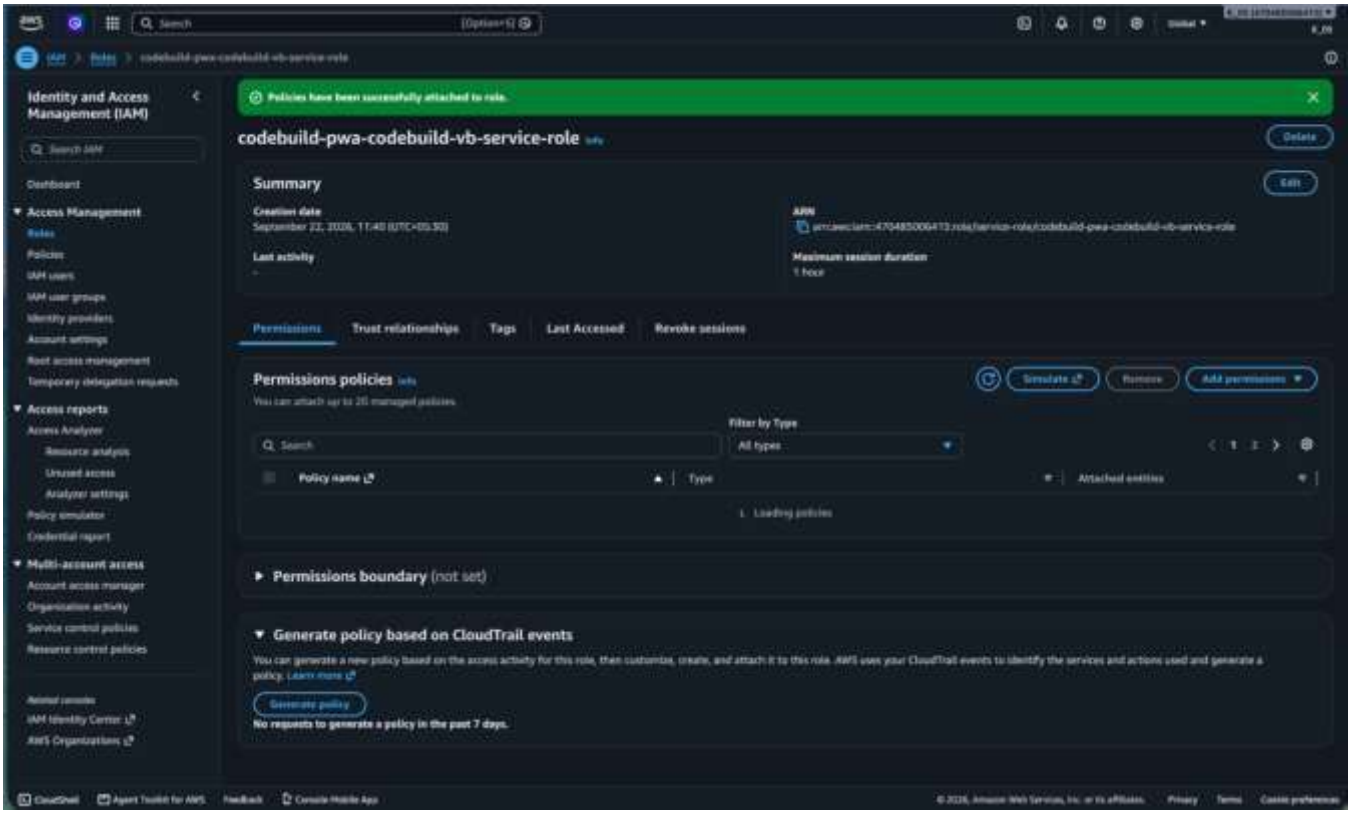


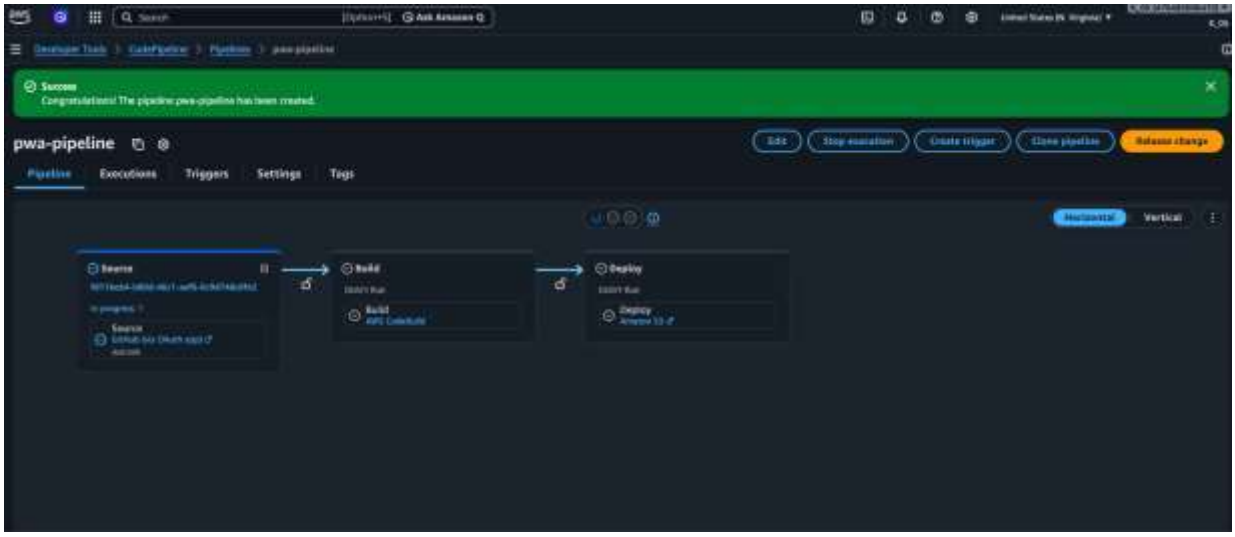
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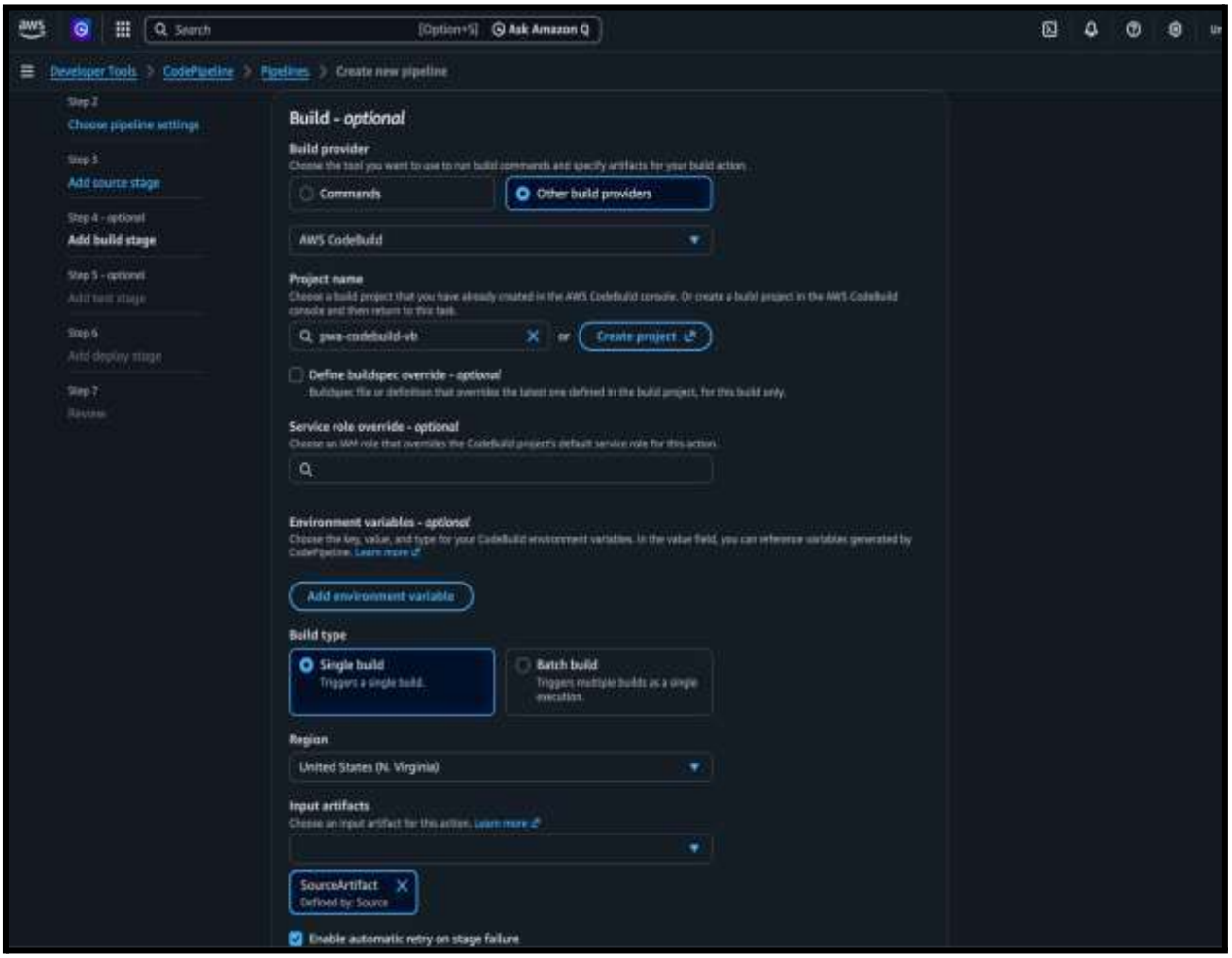


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Conclusion: In this experiment, we successfully studied the implementation of a CI/CD pipeline to host a website on AWS. By using services such as AWS CodePipeline, CodeBuild, and Amazon S3 or EC2, the process of building, testing, and deploying a website can be automated. This reduces manual effort, minimizes errors, and ensures faster and more reliable deployment. Thus, the experiment demonstrates the importance of CI/CD in DevOps for efficient website hosting and continuous updates.

LO4, LO6 mapped successfully.